Worksheet 1.4 — Slope of Tangent

Grade 12 Calculus and Vectors (MCV4U) · Unit 1: Rates of Change and Limits

The Big Ideas

Now that limits are exact (Lessons 1.2–1.3), find the tangent slope at a specific point $x = a$ exactly: $m = \lim_{h\to 0} \dfrac{f(a+h) - f(a)}{h}$. Then the tangent line is $y - f(a) = m(x - a)$.

- $m = \lim_{h\to 0} \dfrac{f(a+h) - f(a)}{h}$ — expand, simplify to cancel h, then let $h \to 0$.
- Tangent line at a: $y - f(a) = m(x - a)$.
- This is the exact version of Lesson 1.1's estimate.

The Big Picture

Exact, not estimated

Lesson 1.1 estimated the tangent slope with a tiny h. Now that limits are exact, the tangent slope at $x = a$ is $m = \lim_{h\to 0} \dfrac{f(a+h) - f(a)}{h}$, computed precisely.

One point at a time

This limit is evaluated at a specific number a the next lesson generalizes it to every x at once.

The tangent line

Once m and the point $(a, f(a))$ are known, the tangent line follows from point-slope form: $y - f(a) = m(x - a)$.

Worked Examples

Example 1 — Confirming the estimate

Find the exact tangent slope of $f(x) = 3x^2$ at $x = 2$.

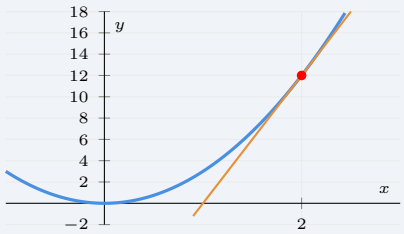
Solution. **Step 1 (form the quotient):** $\dfrac{f(2+h) - f(2)}{h} = \dfrac{3(2+h)^2 - 12}{h}$.

Step 2 (expand): $3(2+h)^2 = 3(4 + 4h + h^2) = 12 + 12h + 3h^2$, so the numerator is $12h + 3h^2$.

Step 3 (factor and cancel h): $\dfrac{12h + 3h^2}{h} = 12 + 3h$.

Step 4 (limit): $\lim_{h\to 0}(12 + 3h) = 12$.

Conclusion: $m = 12$ exactly — matching Lesson 1.1's estimate of ≈ 12:



Example 2 — Tangent line

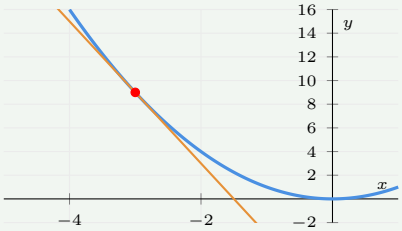
Find the tangent line to $y = x^2$ at $x = -3$.

Solution. **Step 1 (find the slope):** $\dfrac{(-3+h)^2 - 9}{h} = \dfrac{-6h + h^2}{h} = -6 + h \to -6.$

Step 2 (find the point): $f(-3) = 9$, so the point is $(-3, 9)$.

Step 3 (point-slope form): $y - 9 = -6(x + 3)$.

Conclusion: $y = -6x - 9$:



Example 3 — Polynomial

Find the exact tangent slope of $f(x) = 2x^2 + x$ at $x = -1$.

Solution. **Step 1 (form the quotient):** $\dfrac{f(-1+h) - f(-1)}{h} = \dfrac{[2(-1+h)^2 + (-1+h)] - 1}{h}.$

Step 2 (expand): $2(-1+h)^2 + (-1+h) = 2(1-2h+h^2) - 1 + h = 2 - 4h + 2h^2 - 1 + h = 1 - 3h + 2h^2$, so the numerator is $-3h + 2h^2$.

Step 3 (factor and cancel h): $\dfrac{-3h + 2h^2}{h} = -3 + 2h.$

Step 4 (limit): $\lim_{h \to 0}(-3 + 2h) = -3.$

Conclusion: $m = -3$.

Example 4 — Reciprocal

Find the exact tangent slope of $f(x) = \dfrac{1}{x}$ at $x = 4$.

Solution. **Step 1 (form the quotient):** $\dfrac{f(4+h) - f(4)}{h} = \dfrac{\frac{1}{4+h} - \frac{1}{4}}{h}.$

Step 2 (combine the fractions): $\dfrac{1}{4+h} - \dfrac{1}{4} = \dfrac{4 - (4+h)}{4(4+h)} = \dfrac{-h}{4(4+h)}.$

Step 3 (divide by h and cancel): $\dfrac{-h}{4(4+h)h} = \dfrac{-1}{4(4+h)}.$

Step 4 (limit): $\lim_{h \to 0} \dfrac{-1}{4(4+h)} = \dfrac{-1}{16}.$

Conclusion: $m = -\dfrac{1}{16}.$

Example 5 — Radical

Find the exact tangent slope of $f(x) = \sqrt{x}$ at $x = 16$.

Solution. **Step 1 (form the quotient):** $\dfrac{\sqrt{16+h}-\sqrt{16}}{h} = \dfrac{\sqrt{16+h}-4}{h}$.

Step 2 (multiply by the conjugate): $\dfrac{(16+h)-16}{h(\sqrt{16+h}+4)} = \dfrac{h}{h(\sqrt{16+h}+4)}$.

Step 3 (cancel h): $= \dfrac{1}{\sqrt{16+h}+4}$.

Step 4 (limit): $\lim_{h\to 0} \dfrac{1}{\sqrt{16+h}+4} = \dfrac{1}{4+4}$.

Conclusion: $m = \dfrac{1}{8}$.

Example 6 — Polynomial

Find the exact tangent slope of $f(x) = x^2 + 5x$ at $x = 2$.

Solution. **Step 1 (form the quotient):** $\dfrac{[(2+h)^2+5(2+h)]-14}{h}$.

Step 2 (expand): $(2+h)^2 + 5(2+h) = 4 + 4h + h^2 + 10 + 5h = 14 + 9h + h^2$, so the numerator is $9h + h^2$.

Step 3 (factor and cancel h): $\dfrac{9h+h^2}{h} = 9 + h$.

Step 4 (limit): $\lim_{h\to 0}(9+h) = 9$.

Conclusion: $m = 9$.

Example 7 — Tangent line

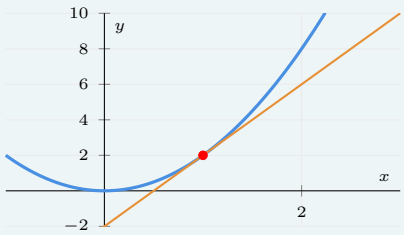
Find the tangent line to $y = 2x^2$ at $x = 1$.

Solution. **Step 1 (find the slope):** $\dfrac{2(1+h)^2-2}{h} = \dfrac{4h+2h^2}{h} = 4 + 2h \to 4$.

Step 2 (find the point): $f(1) = 2$, so the point is $(1, 2)$.

Step 3 (point-slope form): $y - 2 = 4(x - 1)$.

Conclusion: $y = 4x - 2$:



Example 8 — Linear

Find the exact tangent slope of $f(x) = 5x - 3$ at $x = 4$.

Solution. **Step 1 (form the quotient):** $\dfrac{[5(4+h)-3]-17}{h}$.

Step 2 (expand): $5(4+h)-3 = 20+5h-3 = 17+5h$, so the numerator is $5h$.

Step 3 (divide by h): $\dfrac{5h}{h} = 5$ — no h remains.

Conclusion: $m = 5$ (a line's slope is constant everywhere).

Example 9 — Polynomial

Find the exact tangent slope of $f(x) = 4x^2 - x$ at $x = -1$.

Solution. **Step 1 (form the quotient):** $\dfrac{[4(-1+h)^2-(-1+h)]-5}{h}$.

Step 2 (expand): $4(-1+h)^2-(-1+h) = 4(1-2h+h^2)+1-h = 4-8h+4h^2+1-h = 5-9h+4h^2$, so the numerator is $-9h+4h^2$.

Step 3 (factor and cancel h): $\dfrac{-9h+4h^2}{h} = -9+4h$.

Step 4 (limit): $\lim_{h\to 0}(-9+4h) = -9$.

Conclusion: $m = -9$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Find the exact tangent slope of $f(x) = x^2$ at $x = 7$.

Question 2

Find the tangent line to $y = x^2$ at $x = -4$.

Question 3

Find the exact tangent slope of $f(x) = 2x^2$ at $x = 4$.

Question 4

Find the exact tangent slope of $f(x) = \dfrac{1}{x}$ at $x = -3$.

Question 5

Find the exact tangent slope of $f(x) = \sqrt{x}$ at $x = 9$.

Question 6

Find the exact tangent slope of $f(x) = 3x^2 - x$ at $x = 2$.

Question 7

Find the tangent line to $y = x^2$ at $x = -2$.

Question 8

Find the exact tangent slope of $f(x) = 5x - 3$ at $x = 10$.

Question 9

Find the exact tangent slope of $f(x) = x^2 + 5x$ at $x = 0$.

Question 10

Find the exact tangent slope of $f(x) = 4x^2 - x$ at $x = 1$.

Question 11

Find the tangent line to $y = 3x^2$ at $x = 1$.

Question 12

Find the exact tangent slope of $f(x) = \dfrac{1}{x}$ at $x = -1$.

Question 13 — Challenge

Find the exact tangent slope of $f(x) = \sqrt{x}$ at $x = 1$ (rationalize).

Question Recap & Answer Key

Question Recap

1. Find the exact tangent slope of $f(x) = x^2$ at $x = 7$.
2. Find the tangent line to $y = x^2$ at $x = -4$.
3. Find the exact tangent slope of $f(x) = 2x^2$ at $x = 4$.
4. Find the exact tangent slope of $f(x) = \dfrac{1}{x}$ at $x = -3$.
5. Find the exact tangent slope of $f(x) = \sqrt{x}$ at $x = 9$.
6. Find the exact tangent slope of $f(x) = 3x^2 - x$ at $x = 2$.
7. Find the tangent line to $y = x^2$ at $x = -2$.
8. Find the exact tangent slope of $f(x) = 5x - 3$ at $x = 10$.
9. Find the exact tangent slope of $f(x) = x^2 + 5x$ at $x = 0$.
10. Find the exact tangent slope of $f(x) = 4x^2 - x$ at $x = 1$.
11. Find the tangent line to $y = 3x^2$ at $x = 1$.
12. Find the exact tangent slope of $f(x) = \dfrac{1}{x}$ at $x = -1$.
13. Find the exact tangent slope of $f(x) = \sqrt{x}$ at $x = 1$ (rationalize).

✓ Answer Key

1. 14
2. $y = -8x - 16$
3. 16
4. $-\frac{1}{9}$
5. $\frac{1}{6}$
6. 11
7. $y = -4x - 4$
8. 5
9. 5
10. 7
11. $y = 6x - 3$
12. -1
13. $\frac{1}{2}$